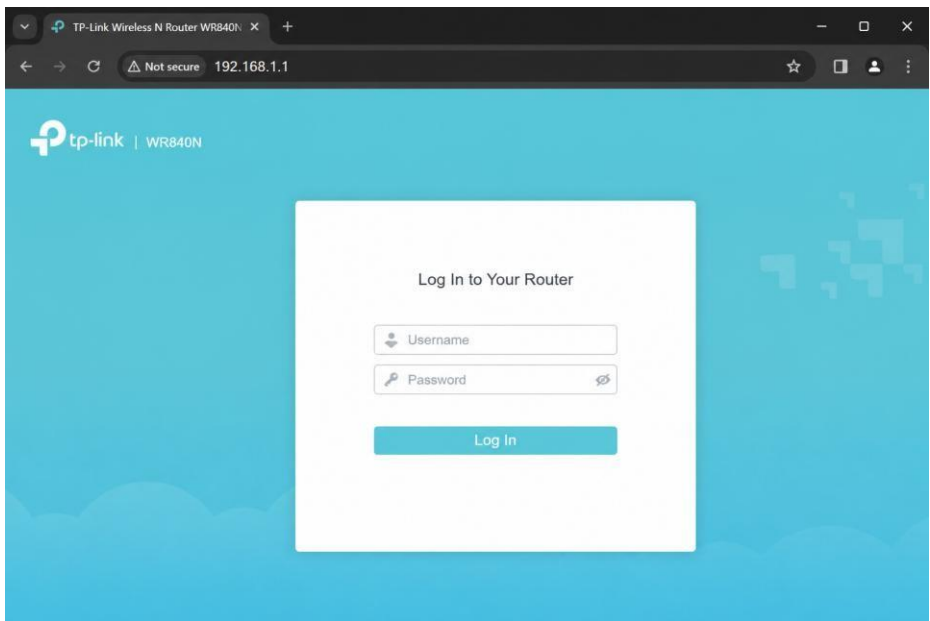


Configuring Basic Network Hardware Theory

- 1) Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



- 2) Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Parameter	Details
IP Address	192.168.1.105
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

- 3) Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

✚ **Before Change (DHCP):**

- IP Address: Automatically Assigned
- Router/Gateway: 192.168.1.1 ✚

Static IP Settings Applied:

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 8.8.8.8

✚ **Testing Result:**

After assigning the static IP address, the device was able to connect to the internet successfully. Websites opened normally, which confirmed that the static IP configuration was working.

✚ **After Testing:**

The IP settings were changed back to Obtain IP Address Automatically (DHCP) to prevent future network problems.

4) In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

- If two devices are assigned the same IP address, an IP address conflict occurs. One or both devices may lose network or internet access, causing connection problems until each device has a unique IP address.

